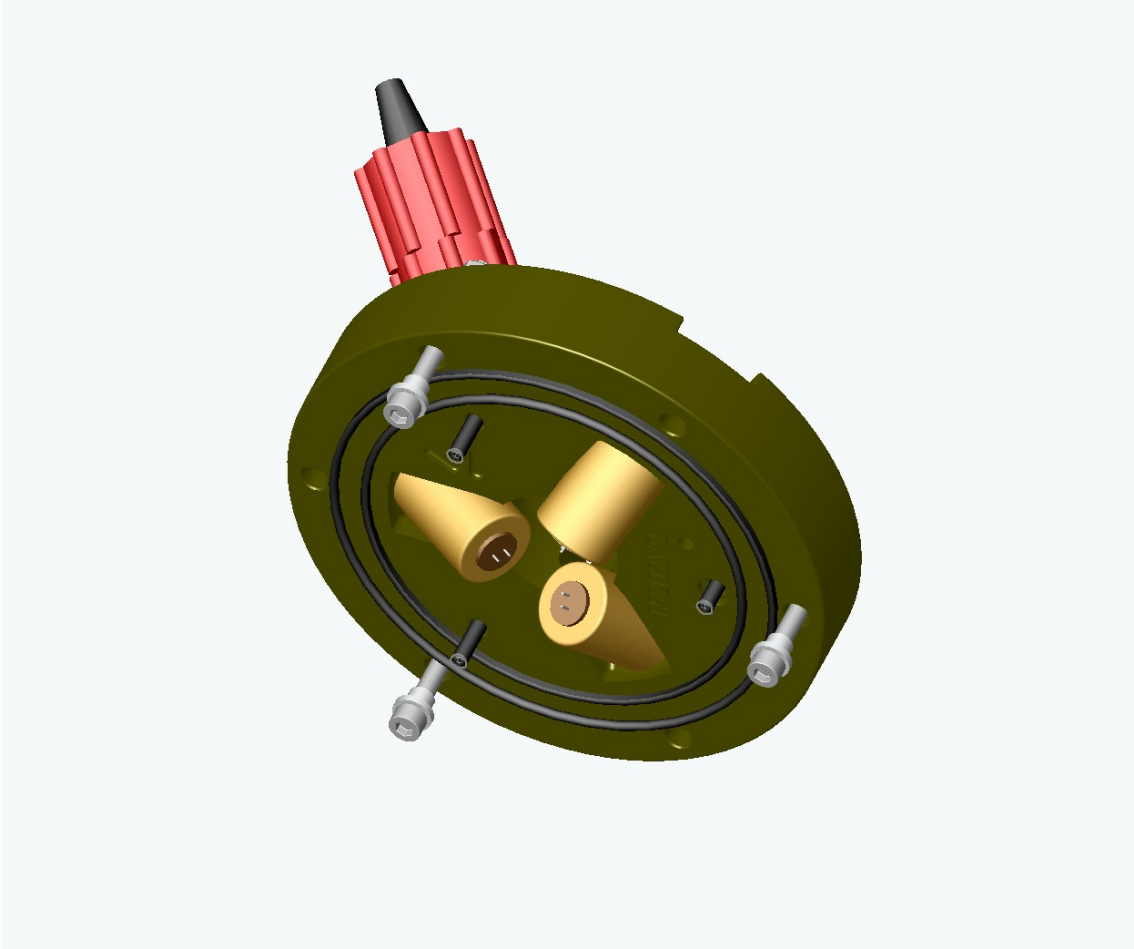
CARRACK MEASUREMENT TECHNOLOGY

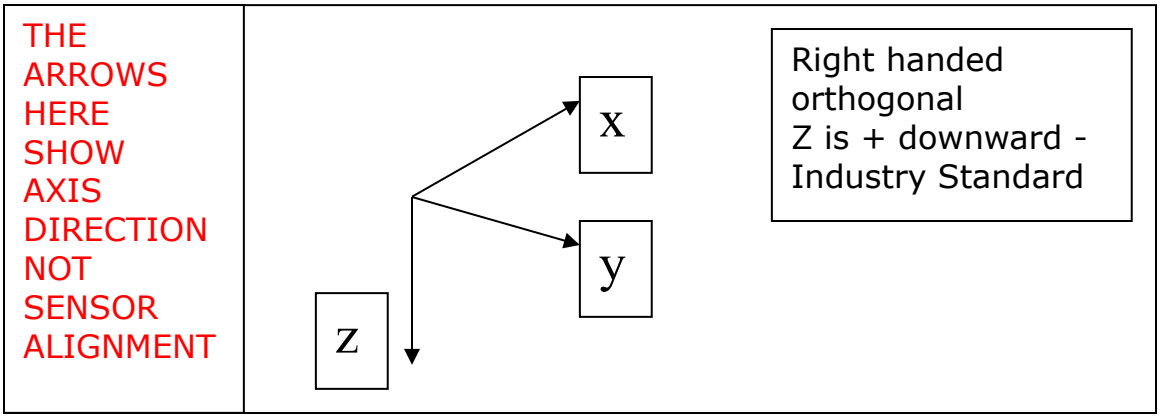
TRILOBIT GEOPHONE CONFIGURATIONS

Tim Owen 19th Feb 2010



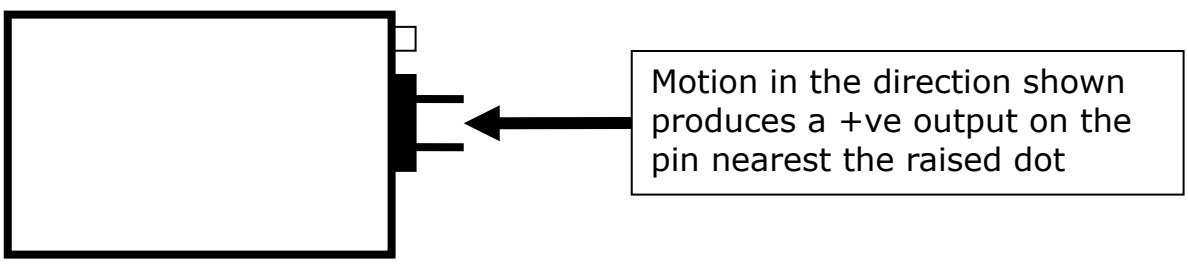
This report looks at the Galperin configuration for the first production Trilobits using the 'inverted' galperin sensor housing and the required conversion matrix.

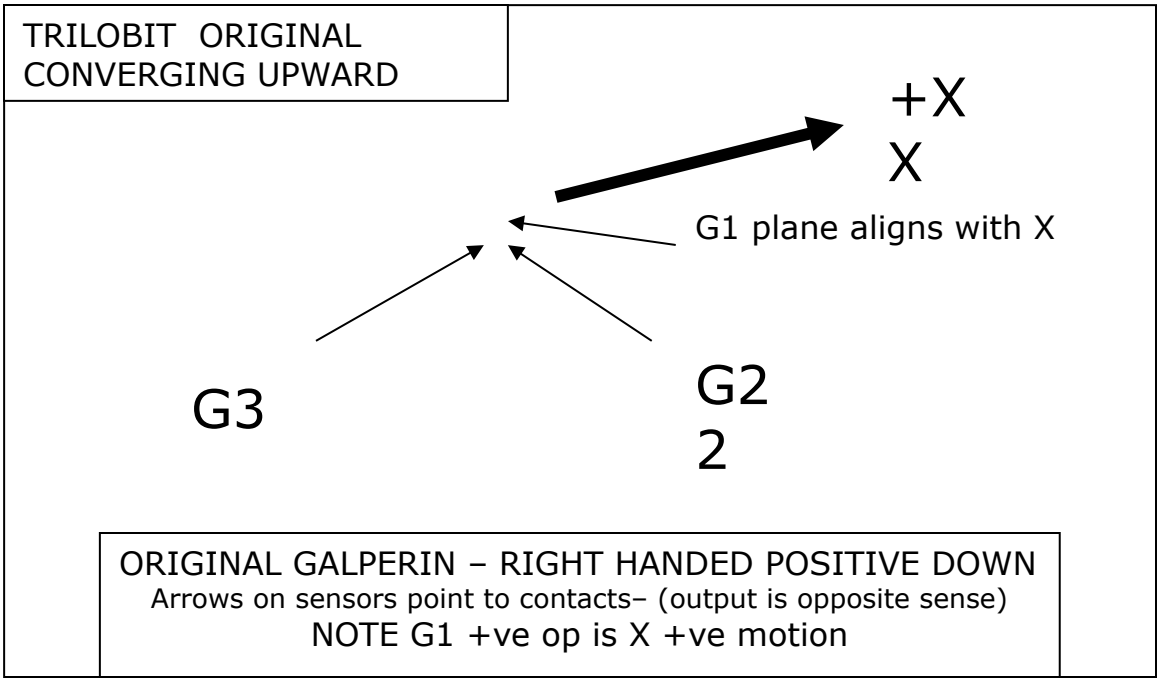
The convention for the representation of seismic motion in the industry is a 3 component orthogonal axis aligned with one vertical and two horizontal components. The axes are arranged as a 'right handed' set – i.e. what you get if you use thumb, first and second fingers of your right hand as x,y and z respectively. The seismic convention is to have z as positive downward – in contrast to many spatial coordinate systems which have z upwards.



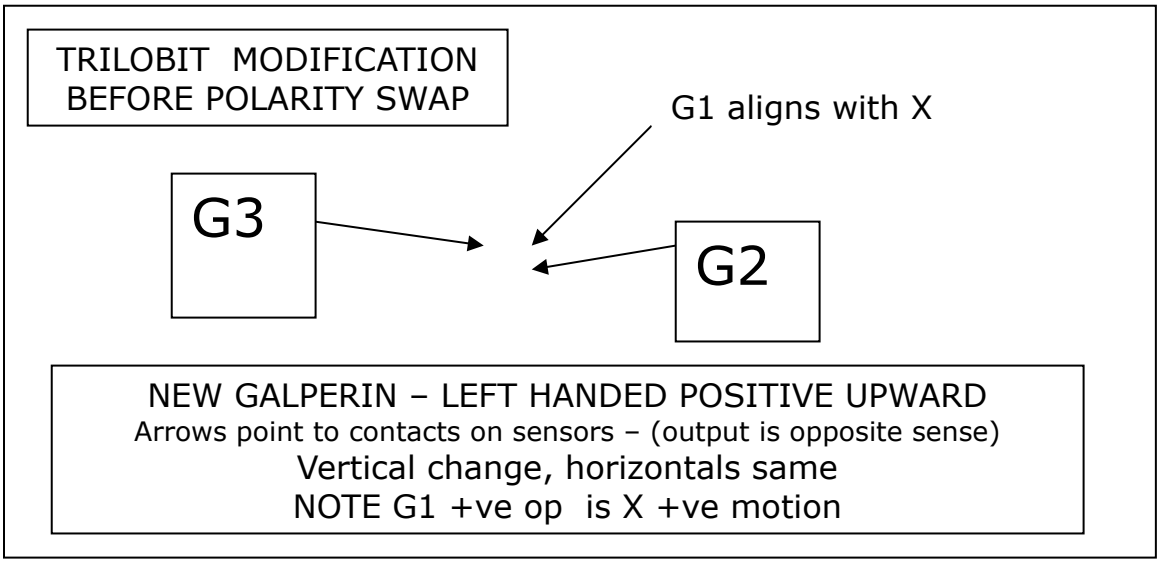
The Trilobit, along with most industry seabed geophone systems, have their sensors arranged in a Galperin configuration, where the 3 sensors are arranged in an orthogonal set, but aligned symmetrically about the vertical axis so that each sensor is inclined at 54.7356 degrees to the vertical and equally spaced in the horizontal plane. There are two possible galperin alignments, sensors converging up or converging down, and the sensors can be arranged as a right handed or left handed set in each case. The polarity of the sensors can, of course, also be reversed, so there are 8 possibilities (some overlap), and no generally accepted convention. Since processing is done in conventional xyz space, the Galperin outputs have to be manipulated during the early stages of processing to convert the signals from the G1,G2 and G3 sensors to x,y and z. Any Galperin arrangement can be converted to any xyz convention in this process. Therefore if the internal arrangement of the geophones is accurate, consistent and known, it can always be converted to the required xyz representation.

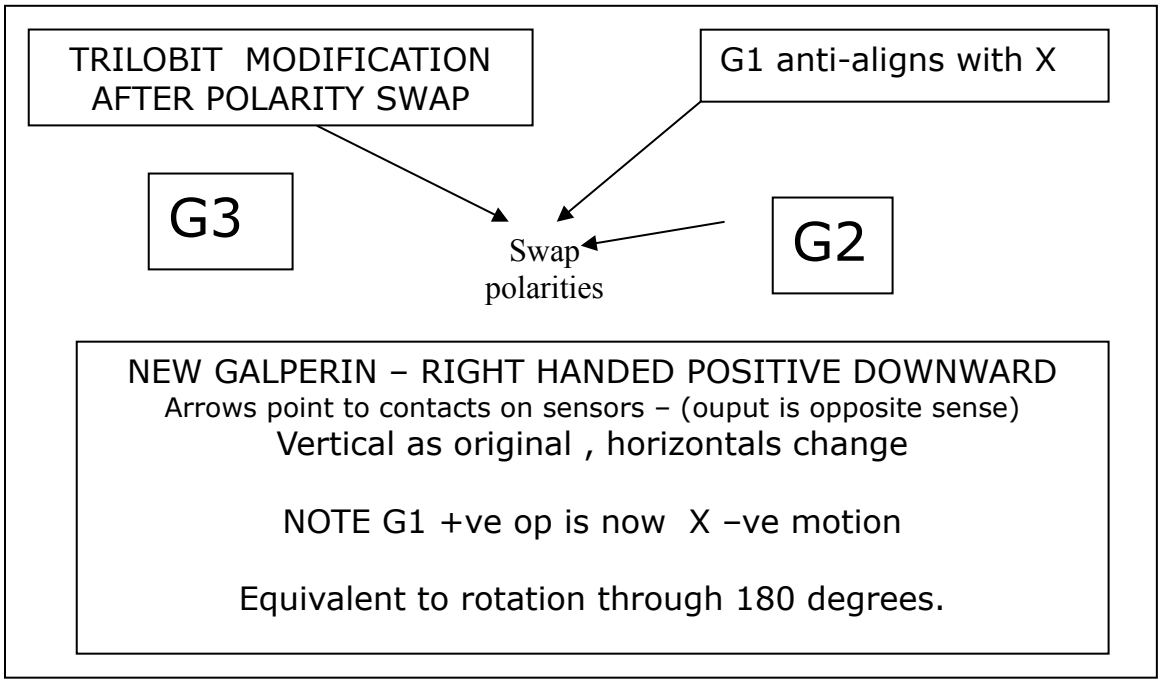
Note that these diagrams use two different conventions:- The axis system above is shown with the arrows pointing in the direction that gives a positive output for positive motion. The sensors below are shown with the arrowhead at the end with the connections – motion in the OPPOSITE direction to the arrow direction produces a positive output. The Sensors have their nominal +ve pin marked by a raised dot, and produce a +ve output for motion away from the connector end towards the blank end





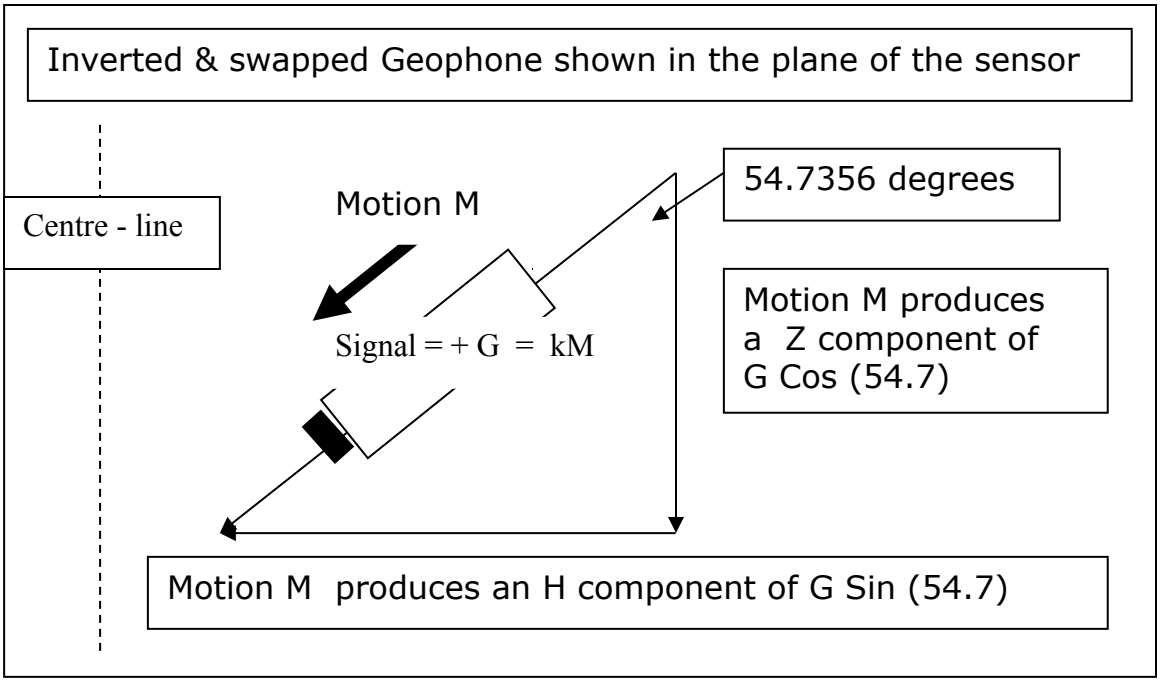
The Trilobit was changed from Galperin converging upward as a right handed set to converging downwards because it theoretically gives a higher output in the presence of rotations, and more practically, it simplified assembly. The change was effected by rotating the geophones in their current positions to point down instead of up. This had the effect of changing the configuration to a left handed set and changing the polarity of the converted vertical (z) component, but not the converted horizontal components. The original orientation had produced positive signals on all geophones for a downward tap – which is the standard first level industry test. The new configuration, if wired with the indicated geophone sense, produced the opposite polarity – counter intuitive for someone used to geophone testing.





Changing the polarity of the sensors while keeping them in their original horizontal positions reverts the arrangement to a right handed set and produces the required test result, but reverses the converted horizontals relative to the original outputs. It would be possible to correct this by rotating the sensor housing by 180 degrees in the horizontal plane, but this is not a convenient modification from a mechanical point of view and is not necessary, since the converting software can be changed easily by inverting the X and Y output.

The conversion is done by adding the contributions of each of the 3 sensors in the x,y and z directions (that is 9 contributions). This can be done in a number of ways, but basically it consists of finding out how much each GI,G2 and G3 output contributes to each x,y and z direction and then adding them in. These can be written as individual sums or, more usually, as a matrix (which in this case is just a way of writing the same thing)



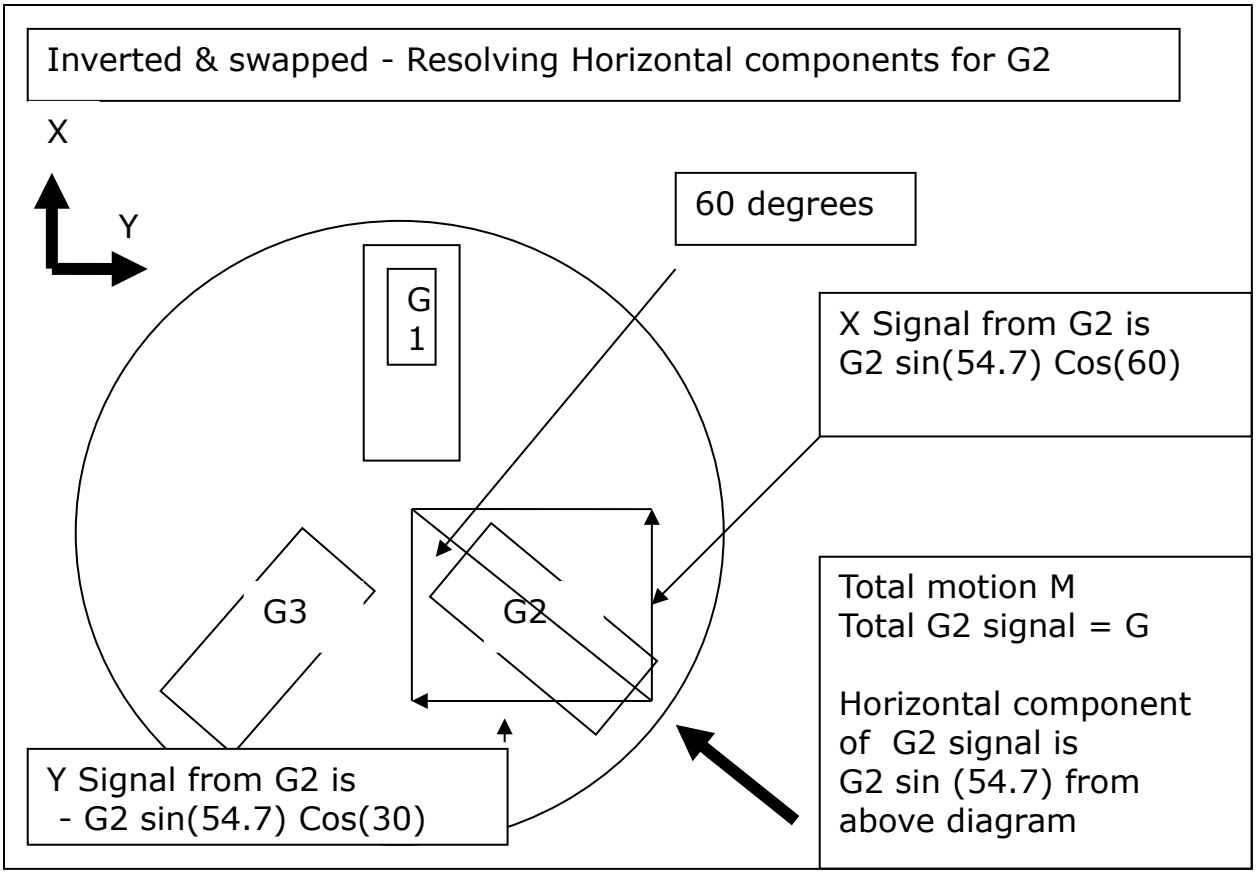
Consider the vertical sum;- G1, G2 and G3 all contribute equally to Z, but Z doesn't align with the sensors. We need to calculate how much the G signals contribute to the Z, as resolved in the diagram above.

So $Z = G1 \cos(54.7) + G2 \cos(54.7) + G3 \cos(54.7) = (G1+G2+G3) \cos(57.4)$ (i)

Since we want a +ve output for downward motion, and that is what our G's give since we flipped them **&** swapped the connections, the sign is correct and the same as the original.

Similarly for the horizontals;-

For X, which we defined to be in the direction of G1, we now have the polarity of the sensors inverted, so we have to change the signs. Only a part (sin(54.7)) of the G outputs is aligned in the horizontal plane, so we have to multiply each G by that factor as well as the geometry in the horizontal plane.



G3 gives similar result except that Y component is + G2 sin(54.7) cos(30)

X = - G1 sin(54.7) + G2 sin (54.7) cos (60) + G3 sin(54.7) cos (60).

Clearly G1 doesn't contribute to the Y output as it's a right angles to Y, so

Y = 0 - G2 sin (54.7) cos (30) + G3 sin (54.7) cos (30).

A 3 x 3 matrix is a convenient way of writing 3 additions with 9 multiplications like this:

$$\begin{matrix} X \\ Y \\ Z \end{matrix} = \begin{matrix} a & b & c \\ d & e & f \\ g & h & I \end{matrix} \begin{matrix} G1 \\ G2 \\ G3 \end{matrix}$$

It is the same as writing

$$X = aG1 + bG2 + cG3$$
$$Y = dG1 + eG2 + fG3$$
$$Z = gG1 + hG2 + iG3$$

So our conversion matrix for converting the new converging downwards Galperin configured as a Right handed set by reversing polarities on the geophone connections into a Right Handed orthogonal coordinate system with Z aligned downwards and X axis aligned with the G1 sensor that has its connections in the –X end is:

$$\begin{vmatrix} X \\ Y \\ Z \end{vmatrix} = \begin{vmatrix} -\sin(54.7) & \sin(54.7)\cos(60) & \sin(54.7)\cos(60) \\ 0 & -\sin(54.7)\cos(30) & \sin(54.7)\cos(30) \\ \cos(54.7) & \cos(54.7) & \cos(54.7) \end{vmatrix} \begin{vmatrix} G1 \\ G2 \\ G3 \end{vmatrix}$$

Instead of using sines and cosines you can substitute the geometric factors to give the same answers - $\sin(54.7) = 2/\sqrt{6}$, $\cos(60) = 1/2$, $\cos(54.7) = \sqrt{2}/\sqrt{6}$ etc.

Note 54.7 is more accurately 54.736 degrees.

I have verified my conversion matrix by tap tests on an original geophone and on T5200V2 SN 1226 which is a new configuration downwards converging geophone WITHOUT the sensor connections swapped using a Picoscope at CGGV Bergen.

Tap test result (tap direction as below) :-

ORIGINAL

TAP	Dirn	G1	G2	G3
1	+Z	+	+	+
2	+X	+	-	-
3	+Y	0	+	-
4	G3	0	0	-
5	G2	0	-	0

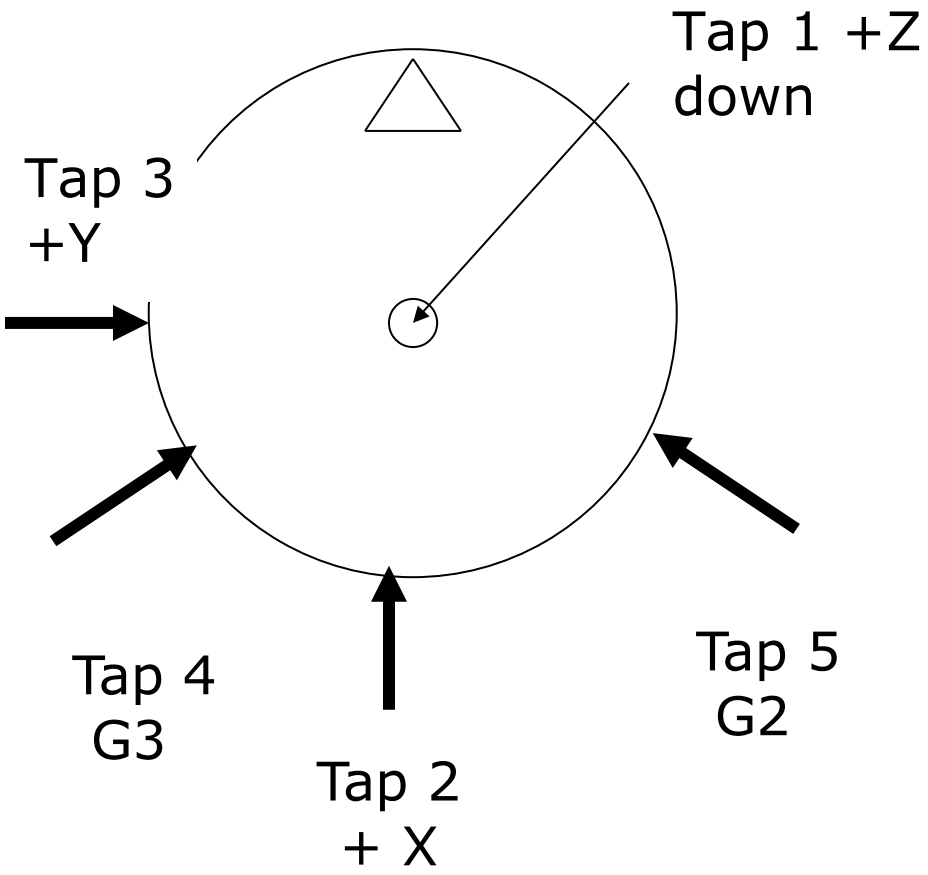
DOWNWARD, NOT POLARITY SWAPPED
SENSOR T5200v2 SN1226

TAP	Dirn	G1	G2	G3
1	+Z	-	-	-
2	+X	+	-	-
3	+Y	0	+	-
4	G3	0	0	-
5	G2	0	-	0

Inverting all outputs of the T5200V2 sensor corresponds to my conversion matrix for polarity reversed T5200V2 geophones.

TAP TEST OF CONVERGING DOWNWARD SWAPPED POLARITY GEOPHONE T5200V2 SWAPPED AS PROPOSED FEB 2010

TOP VIEW OF GEOPHONE.



TAP	Dirn	G1	G2	G3
1	+Z	+	+	+
2	+X	--	+	+
3	+Y	0	-	+
4	G3	0	0	+
5	G2	0	+	0